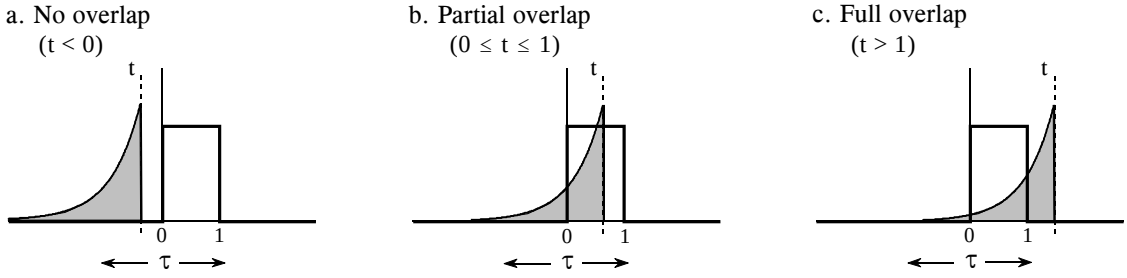

FIGURE 13-6
Calculating a convolution by segments. Since many continuous signals are defined by *regions*, the convolution calculation must be performed region-by-region. In this example, calculation of the output signal is broken into three sections: (a) no overlap, (b) partial overlap, and (c) total overlap, of the input signal and the shifted-flipped impulse response.

mathematical expression. This is very common in continuous signal processing. It is usually essential to draw a picture of how the two signals shift over each other for various values of t. In this example, Fig. 13-6a shows that the two signals do not overlap at all for $t < 0$. This means that the product of the two signals is zero at all locations along the τ axis, and the resulting output signal is:

$$y(t) = 0 \qquad \text{for } t < 0$$

A second case is illustrated in (b), where t is between 0 and 1. Here the two signals partially overlap, resulting in their product having nonzero values between $\tau = 0$ and $\tau = t$. Since this is the only nonzero region, it is the only section where the integral needs to be evaluated. This provides the output signal for $0 \le t \le 1$, given by:

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau \qquad \text{(start with Eq. 13-1)}$$

$$y(t) = \int_{0}^{t} 1 \cdot \alpha e^{-\alpha(t-\tau)}\, d\tau \qquad \text{(plug in the signals)}$$

$$y(t) = e^{-\alpha t}\, [\, e^{\alpha\tau} \,]\Big|_{0}^{t} \qquad \text{(evaluate the integral)}$$

$$y(t) = e^{-\alpha t}\, [\, e^{\alpha t} - 1 \,] \qquad \text{(reduce)}$$

$$y(t) = 1 - e^{-\alpha t} \qquad \text{for } 0 \le t \le 1$$